Day-48

ODATA Adapter.

* OData → open Data protocol
  → What is the actual use of ODATA
  → OData is also developed on Rest Based API
  → Advance Capabilities → OData (Rest API)
  But Standard Rest API → Don't have Advance Capabilities

→ Advance Capabilities:

Rest Based API

Sort filter Select etc

We cannot Select them directly

eg: northwind API

northwind OData Vs API call Rest Based API

- Required → Fields
  → Filter
  → Sorting

Additional Query Capabilities

OData → available

→ not a Seaperate Concept of Rest it's built on Rest Based API

→ only thing is based on additional Capability we generally prefer with OData

* Let's go with what are all the query capabilities, OData URL !

* Let's discuss of OData URL

→ OData (adapter) → SAP CPI

* OData URI how it will be generated

* URL → blog → odata.org → Documentation

→ OData 2.0 → overview ~~URI~~

→ URI conventions

* Main Concept is Host name & port

* PIXcall → Host and port is mandatory

eg: https (protocol), host, port (common one)

OData URI is seperated sections

* https://host:port/<Service root URL>/Resource (or) Entity

* OData consist of Different resources or entities

eg: Calculator.wsdl → operations → Add
Sub
multiply
Subtraction

* https:// host : port / <serviceRootURL> / <resource> / <entity>
? <query option>

↓

if this exists in URL

Then we will call it as query URL's

* postman example → Placing '?' demo

* Now take example of ODATA API

⇒ https:// services.odata.org / V2 / northwind / northwind.svc / Employees

↓ ↓ ↓

host ServiceRootURI ~~Employees~~ Resource

* NOW the difference of HTTP URI & ODATA URI

Why we need to use ~~http~~ OData URL rather than http?

* SOAP Adapter usase ki → prerequisite → WSDL

File (schema definition → Services has what operations)

* ODATA Adapter → prerequisite → Edmx →

File (we want)

* What is Edmx file → Schema definition

* complete schema → what fields } → edmx file
→ what Entity (schema)

* It's a open source. → there is no schema definition for this.

* So edmx file how it will be generated.

What ever we have Service root URL.

https://——————/$ metadata (schema definition file)
Service URL

* demo showing how to copy edmx file of northwind API

* Entity Container → List of Entities

* Employee(Entity) → list of fields.

* order details → fields

⇒ Now query options

OData → RestAPI → Additional capabilities

Query options: ① Select ⑧ format
② expand.
③ filter.
④ ~~sort~~ orderby
⑤ top
⑥ skip
⑦ inline count

* These are Standard Query options

① Select: If you want to call a OData API the Result is getting Complete Records or all the data

Requirement: If you want to get 2 field's on but the it has 40 field's ?

OData is useful but ~~Rest API~~ (HTTP) ↓ not recommended

* OrderID & ProductID

only two field's required → then we use Select Query

?$ ⟶ assume Standard query options

/Entity namely ?$ select = OrderID, ProductID

→ Select Query → It is used to Select only the required field's

* here we need to remember about the custom queries and Standard Queries.

Custom queries	Standard queries
no '$' → symbol	$ → yes
?$	?

* Let's talk about Sorting option

→ Sort the data → Sort the data using ProductID

Eg: 11, 42, 72, 14, 12)

→ I want to sort Based on product ID

$ orderby = productID desc

↓

here default will be asec

* Now about "orderID" → sorting (desc)

$ orderby = orderID desc

* Sort the data → Sequence order

we will use

$ orderby =

* "inlinecount" = the count of the number of Records of that particular API

↳ standard query option

- - - - - { $inlinecount = allpages (or) none

↓

Query options

<m:count>2155</m:count>

↳ Records available

* ~~To go~~ Pagination

* format → v2 - v4 (oData)

↳ Support's → xml)

?$ format = json ↵

↳ used

* top & skip → most important topics ↵

pagination → oData → all records we don't get

→ we get some Records.

* SOAP → heavy weight ↵

SOAP API → call → 10,000 records → will be fetch → will fetch all records

oData / light weight ↵

→ 10000 → single call → 100,200 1000 → only provides result

* Timedelay →

Pagination Concept →

100 / 200
1st page / 2nd page - - - ?

1st page → 1 to 100
2nd page → 101 to 300

* google Search demo → Actions → Response
at end we will see "pages"

* Api call → pagination Quick Respons →

* top → to fetch top number of Records

The top number of Records →

?$top = 200,

* multiple query options → f

?$top = 200 & f skip = 50

Remember: Standard queryoptions ?$

Select → Used to select only required fields
Orderby → used to sort the data (asc/desc)
top → to fetch the top number of record
Skip → Skip the unwanted number of record

inlineCount → To get all the records available in that API

format → To get the result in required format

Syntaxes:

Select: ----/Entityname?$select=Column Name
host: port/ServiceURL/
↓ Standard query
Product ID, orders

Orderby: ---------?$orderby column name asc/desc

top: ---------?$top=200.

Skip: ---------?$top=200&skip=100

Inlinecount ---------?$inlinecount=allpages/none

format: ---------?$format=json

Expand - }→ pending
filter = }